

Suraj Kumar

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TECHNICAL SKILLS

Languages: Python, JavaScript, C++, SQL, Bash

AI & LLM: LangGraph, LangChain, RAG Pipelines, OpenAI APIs, Vector Databases, Prompt Engineering, Ollama

Backend Frameworks: Django, FastAPI, Django REST Framework (DRF), REST APIs, Flask, JWT

Architecture & Messaging: Distributed Systems, Microservices, RabbitMQ, Celery, Redis

Cloud & DevOps: AWS (Lambda, API Gateway, EventBridge, DynamoDB, S3), Docker

Databases: PostgreSQL, MySQL, MongoDB

WORK EXPERIENCE

Brain of Code — *Software Development Intern*

Oct 2025 – Apr 2026

- **Developed** and maintained backend modules using **Django & DRF**, building **REST APIs** and integrating **Redis** caching, reducing average response times by **40%** for high-traffic endpoints.
- **Fixed 10+** critical bugs and optimized existing **PostgreSQL** queries, cutting average query execution time by **20–25%** and improving system stability.

Smart India Hackathon (SIH) — *Backend Developer*

Aug – Sep 2025

- **Engineered** a cross-platform internship application (**Flutter + Flask**) with **MySQL** schemas, **REST APIs**, and **JWT authentication**, connecting **1000+** students with **500+** companies.

OPEN SOURCE CONTRIBUTIONS

Django-CRM (Python/Django Contributor)

[GitHub](#)

- **Resolved** a **data integrity bug** in Lead-to-Request conversion, verified against a **194-test suite** (*Django ORM*).

ai-hub (Python Contributor)

[GitHub](#)

- **Implemented** a **Claude CLI Provider** module by extending the **CLIBridge** architecture with zero core-code changes across **5 files** (+221/-4 lines), resolving **5 maintainer review rounds** before merge (*Python*).

Microsoft AI-For-Beginners (Python Contributor)

[GitHub](#)

- **Strengthened** error handling in image validation functions of Microsoft's AI-For-Beginners (**58k+** **GitHub stars**), replacing unsafe bare **except** clauses per **PEP 8** (*Python*).

13+ open-source contributions across various repositories, with 10 pull requests merged.

KEY PROJECTS

AI-Powered Personal Mental Health Assistant

[Live Demo](#)

- **Designed** a **self-hosted multimodal system** accepting **text, voice, and video** as input, trained on **1L+ data samples** to detect **28 emotions** at **90–94% accuracy**, using **Python**, a **RAG pipeline** with **Vector Databases**, and **LangChain** to ground and orchestrate context-aware responses.
- **Fine-tuned** a local **LLM via Ollama** across **4–5 iterations** with **prompt engineering** for empathetic replies, deployed via **Flask REST APIs** and validated by **50+ test users** for real-time **crisis detection**.

Hospital Management System

[GitHub](#)

- **Architected** a **desktop application** in **Python** with **role-based login** for 3 user roles, managing **5+ modules** (patient records, appointments, billing) using a **MySQL** database with optimized **SQL** queries, and **Bash** scripts to automate setup and backups.

EDUCATION

Babasaheb Bhimrao Ambedkar University

2024–2026

Master of Computer Applications (**MCA**) – Computer Science

Focus: Advanced Algorithms, Cloud Computing, Artificial Intelligence

CERTIFICATIONS & ACHIEVEMENTS

- **Honorable Mention:** Recognized in the **Microsoft Pygame Hackathon** for innovative game logic and design.
- **Competitive Programming:** Solved **250+ Data Structures and Algorithms problems** on LeetCode.
- **Certifications:** Oracle – Cloud Infrastructure 2025 Certified Generative **AI Professional** — Microsoft – **Data Analyst 101 Certified** — Deloitte – **Cybersecurity Job Simulation**